STATISTICAL ANALYSIS AND PERSONALIZED BOOK RECOMMENDATION SYSTEM BASED ON USER PREFERENCE DATA

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DEPARTMENT OF COMPUTER SCIENCE
M.SC.DATA SCIENCE II YEAR
[2024-2026]
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INTRODUCTION

- In today's digital world, millions of books are available on online platforms and digital libraries.
- Due to the large volume of books, users often find it difficult to identify books that match their interests.
- Traditional search methods require users to manually explore books, which is time-consuming and inefficient.
- Recommendation systems help users discover relevant books based on their preferences and behavior.
- This project focuses on developing a **Statistical Analysis and Personalized Book Recommendation System** using user preference data.

PROBLEM STATEMENT

- Existing book recommendation systems often use a single technique, which may reduce recommendation accuracy.
- Techniques like collaborative filtering face cold start and data sparsity issues, reducing effectiveness.
- Content-based methods depend on data quality, which may be inconsistent or incomplete.
- Large datasets can increase processing time and memory usage.
- Therefore, there is a need for a **hybrid recommendation system** that provides accurate, personalized, and efficient book recommendations.

RESEARCH OBJECTIVES

- **Objective 1: To Develop a Hybrid Recommendation System**

  Combine Collaborative Filtering and Content-Based Filtering to generate more accurate and reliable book recommendations.

- **Objective 2: To Handle Cold Start and Data Sparsity Problems**

  Provide recommendations for new users/books and reduce the impact of limited rating data.

- **Objective 3: To Enhance Personalization Using User Preference Data**

  Analyze user ratings and behavior to generate customized book recommendations.

- **Objective 4: To Improve Scalability and Computational Efficiency**

  Develop an efficient system that can process large-scale book datasets with reduced time and memory usage.

- **Objective 5: To Improve Recommendation Quality and Explainability**

  Evaluate recommendation performance and provide clear explanation for suggested books.

Objective 1: Hybrid Recommendation System

Develop a Hybrid Recommendation System by combining Collaborative Filtering and Content-Based Filtering techniques.

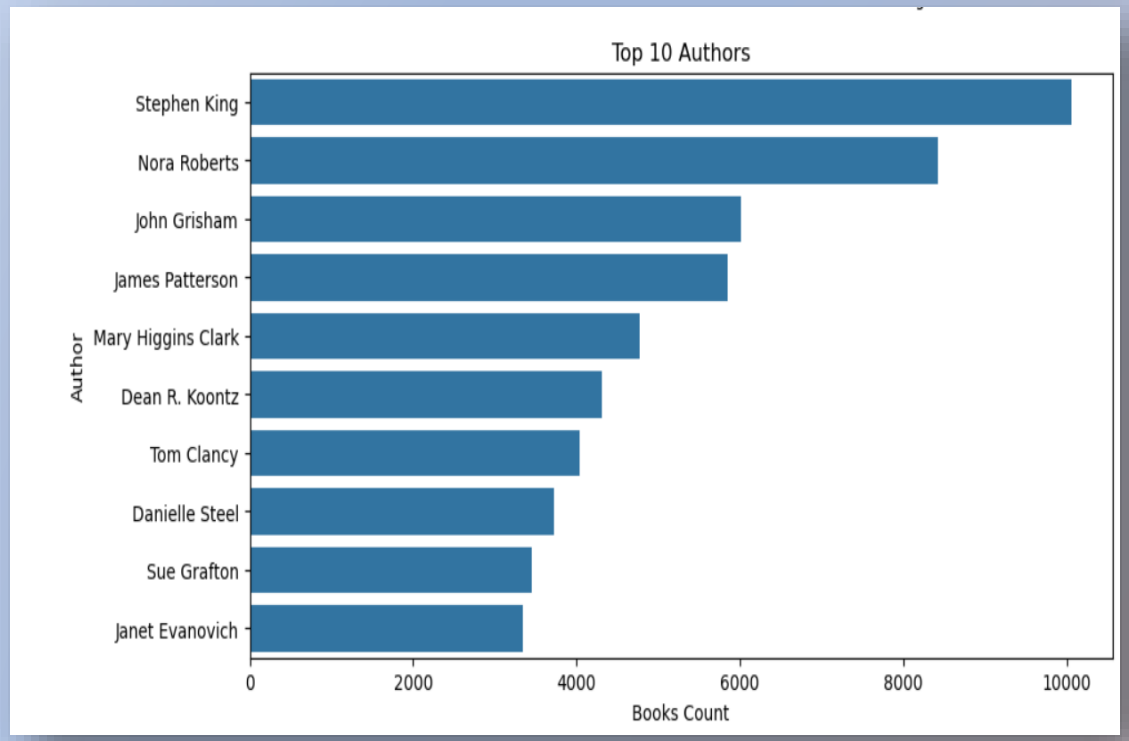
Models Used

- **Collaborative Filtering (SVD):** Predicts user preferences using rating patterns of similar users.
- **Content-Based Filtering** - recommends books similar to those the user already likes.
- **Hybrid Recommendation System -** Combines Collaborative Filtering and Content-Based Filtering to generate more relevant and personalized book recommendations.

Statistical Tools Used

- **TF-IDF Vectorization -** Converts book information (title, author, publisher) into numerical values so that similarity between books can be calculated.
- **Cosine Similarity –** Measures similarity between books. Higher similarity scores indicate more related books.
- **RMSE (Root Mean Square Error) -** Evaluates the prediction error of the recommendation model by measuring the difference between actual ratings and predicted ratings. Lower RMSE indicates better performance.

Visualization Outputs :



Graph 1: Top 10 Authors

Explanation

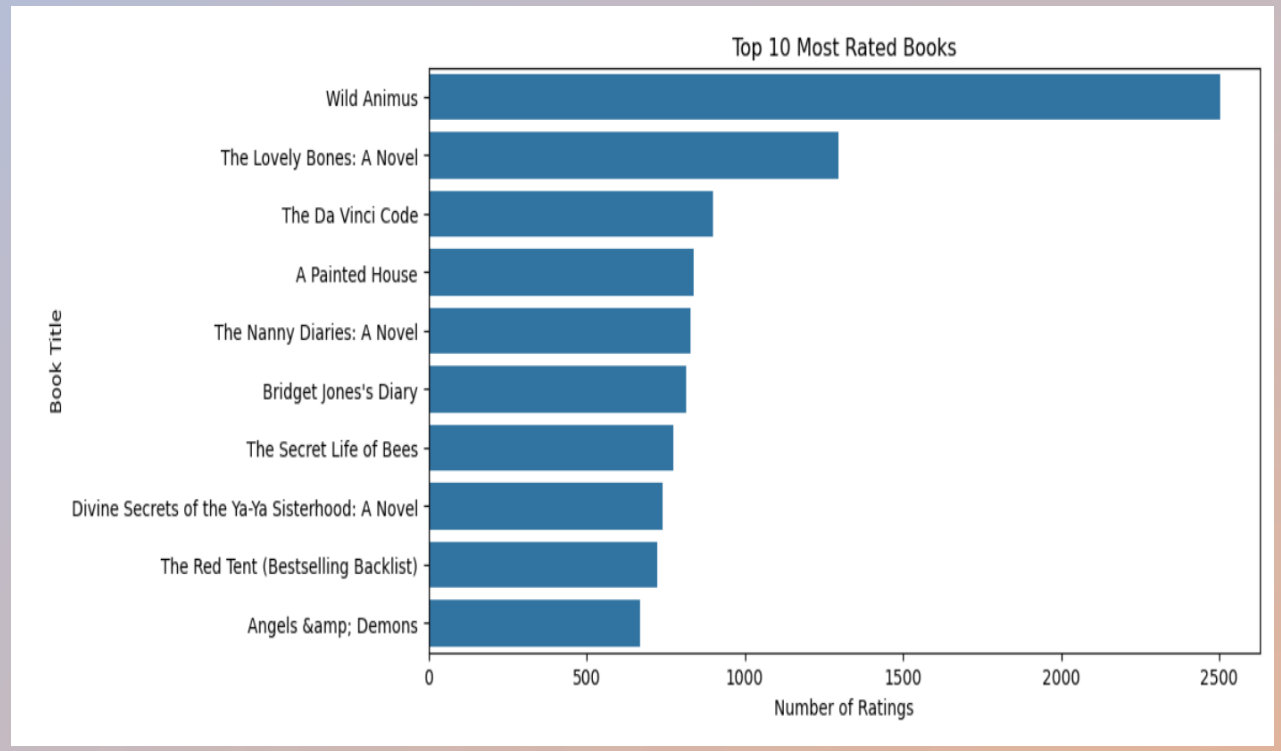
- Shows the most popular authors in the dataset.

Observation

- Certain authors have higher reader engagement.

Result

- Author information helps Content-Based Filtering recommend similar books.



Graph 2: Top 10 Most Rated Books

Explanation

- Shows books with the highest number of user ratings.

Observation

- A few books received significantly more ratings than others.

Result

- These books provide sufficient data for Collaborative Filtering.

Output: Final Hybrid Recommendation System

Rank	Recommended Book	Predicted Rating
1	The Joy Luck Club	3.882
2	The Hundred Secret Senses	3.291
3	The Bonesetter's Daughter	2.661
4	Dragon Tears	2.348
5	Vector	2.298

Output Explanation

The hybrid model generated a ranked list of recommended books along with predicted ratings.

Observation

The books were ranked based on predicted user preference scores, with *The Joy Luck Club* receiving the highest score (3.882).

Result

The Hybrid Recommendation System successfully combined user rating patterns and book similarity information to generate relevant book recommendations.

Objective-2: Handle Cold Start and Data Sparsity Problems

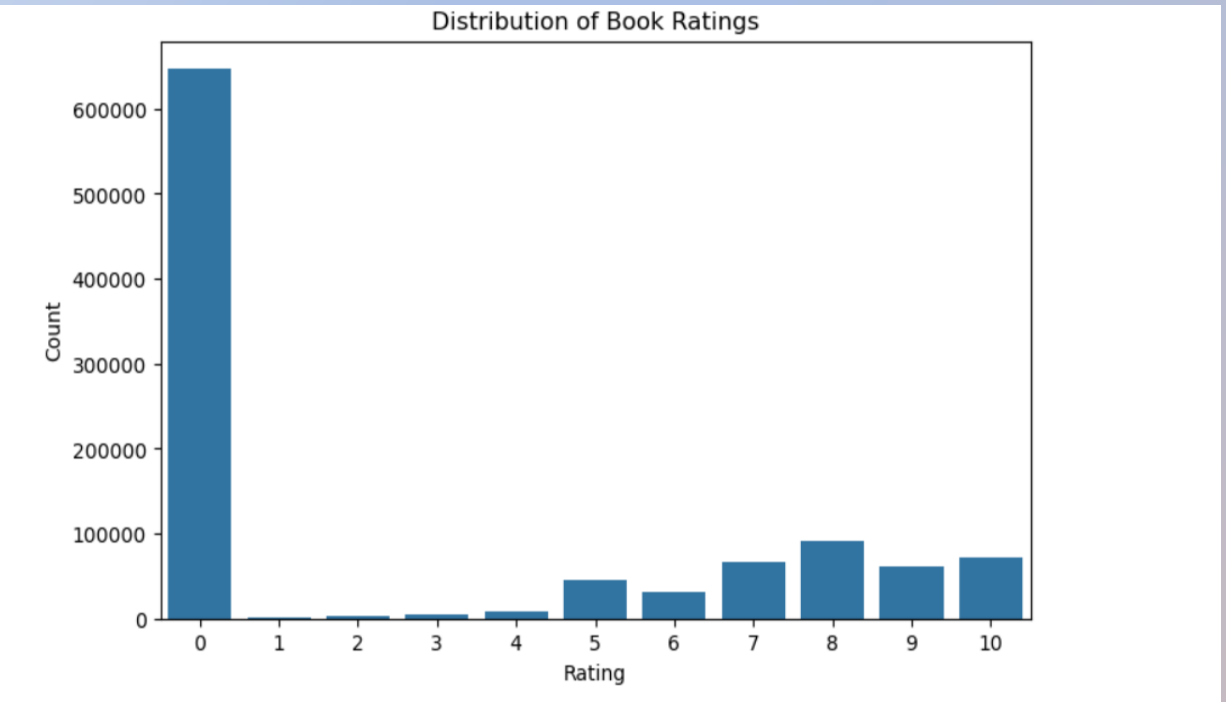
To handle Cold Start and Data Sparsity problems by providing meaningful recommendations when user rating data is limited.

Models Used

- **Hybrid Recommendation System -** Combines Collaborative Filtering and Content-Based Filtering to generate more relevant and personalized book recommendations.
- **Popularity-Based Recommendation -** Recommends highly rated and frequently rated books, especially for new users with no rating history.
- **Collaborative Filtering (SVD) -** Predicts user preferences by analyzing rating patterns of similar users and books.

Statistical Tools Used

- **Rating Count Analysis -** Identifies books with sufficient user ratings.
- **Average Rating Calculation -** Determines overall user preference for books.
- **Descriptive Statistics -** Summarizes the dataset using measures such as mean, median, and standard deviation to understand data characteristics.



Explanation

- This graph shows how ratings are spread across the books in the dataset.
- It helps identify whether books have enough ratings available for generating recommendations.

Observation

- Most books have only a few ratings, while only a small number of books receive many ratings.

Result

- The presence of sparse rating data can affect recommendation accuracy, so popularity-based and hybrid recommendation techniques were used to address this issue.

OUTPUT:

TOP 10 RECOMMENDATIONS FOR NEW USERS

Book Title	Average Rating	Rating Count
Free	8.01	56
The Stand (The Complete and Uncut Edition)	6.17	57
Griffin & Sabine	6.04	72
Harry Potter and the Prisoner of Azkaban	5.85	428
Harry Potter and the Goblet of Fire	5.82	387
The Little Prince	5.81	141
The Cat in the Hat	5.75	53
Harry Potter and the Sorcerer's Stone	5.73	278
The Hobbit	5.70	80
Harry Potter and the Order of the Phoenix	5.50	347

Explanation

This output recommends books for new users based on **Average Rating** and **Rating Count**. To ensure reliable recommendations and reduce data sparsity, only books with **Rating Count ≥ 50** were selected.

Observation

Books such as *Harry Potter*, *The Hobbit*, and *The Little Prince* have high average ratings and a large number of ratings. Books with both high user satisfaction (Average Rating) and sufficient user interactions (Rating Count) were selected.

Result

Cold Start and Data Sparsity issues were reduced by recommending highly rated and frequently rated books, enabling reliable suggestions for new users.

Objective 3: Personalized Book Recommendation System

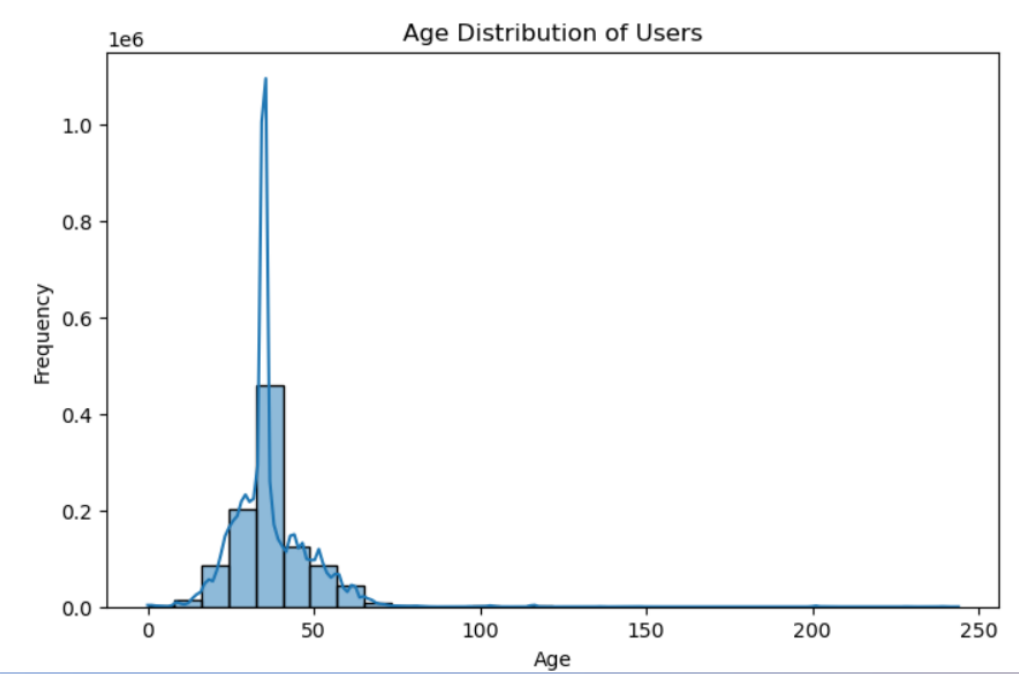
The system recommends books based on user interests, rating behavior, and content similarity, providing personalized suggestions tailored to each user.

Models Used

- **Collaborative Filtering (SVD)** – Predicts user preferences using rating patterns.
- **Content-Based Filtering** – Recommends books similar to those already liked by the us

Statistical Tools Used

- **Independent T-Test** – Compares ratings between two user groups.
- **ANOVA** – Compares ratings across multiple age groups.
- **TF-IDF Vectorization** – Converts book features into numerical vectors.
- **Cosine Similarity** – Measures similarity between books.




```
========== T-TEST ==========

T-Statistic: -1.838897372175017
P-Value: 0.06593056977967114
Result: No significant difference exists.

========== ANOVA TEST ==========

F-Statistic: 287.62838364935726
P-Value: 1.316471911905554e-125
Result: Significant difference among age groups.
```

Explanation:

- X-axis Represents the age of users and Y-axis Represents the number of users (in millions).
- It Shows how users are distributed across different age groups and Helps identify the most common reader age groups in the dataset.

Observation

- Most users are concentrated in the 30–45 years age range, with the highest frequency around 35–40 years. Very few users belong to extremely young or older age groups.

Result

- Young and middle-aged users contribute most ratings, providing enough preference data for accurate personalized recommendations.

Output 1: Sample Prediction

```
Sample Prediction
User ID: 276725
ISBN: 034545104X
Predicted Rating: 1.0008013354014151
```

Explanation

This output shows the predicted rating generated by the SVD model for a specific user-book pair. The model estimates how much User 276725 may like the book with ISBN 034545104X based on previous rating patterns.

Observation

The prediction is generated for a specific user, not for all users. Different users receive different predicted ratings, allowing the system to recommend books according to individual preferences.

Result

The system successfully learned user preferences and can generate personalized recommendations for individual users.

Output 2: Recommended Books

```
================================================
RECOMMENDED BOOKS
================================================
                        Book-Title     Book-Author                Publisher
874   The Hundred Secret Senses          Amy Tan         Putnam Pub Group
872   The Bonesetter's Daughter          Amy Tan  Putnam Publishing Group
22              The Joy Luck Club          Amy Tan     Prentice Hall (K-12)
2803                       Vector       Robin Cook         Putnam Pub Group
1715                Dragon Tears   Dean R. Koontz         Putnam Pub Group
```

Explanation

This output displays the books recommended by the Content-Based Filtering model based on book similarity using features such as title, author, and publisher.

Observation

Most of the recommended books are written by **Amy Tan** or share similar content characteristics and publishing information.

Result

The system successfully generated personalized book recommendations by identifying books similar to the user's selected book.

Objective 4: Improve Scalability and Computational Efficiency

The system improves efficiency by reducing unnecessary data, making recommendations faster and more effective.

Model Used

- **Content-Based Filtering -** Recommends books based on similarity of book features.

Statistical Techniques Used

Dataset Reduction

- Selects only the most relevant books for processing.

Duplicate Removal

- Removes repeated book titles to reduce redundancy.

TF-IDF Matrix Optimization

- Converts book information into an efficient numerical format for faster computation.

Output

```
Books Used For Content-Based Filtering:
(5000, 8)

After Removing Duplicate Titles:
(4469, 8)

Feature Column Created Successfully!

TF-IDF Matrix Shape:
(4469, 6446)

Index Mapping Created Successfully!
```

Output Explanation

- Dataset size was reduced from 5000 to 4469 books by removing duplicate titles.
- TF-IDF converted the books into 6446 unique content features for recommendation processing.

Observation

- Duplicate books were successfully removed, reducing unnecessary data.
- A total of 6446 unique words/features were extracted from book information.

Result

- Less data was used, making the system faster.
- The system generates the recommendation more efficiently.

Objective 5: Evaluate Recommendation Quality

The recommendation system is evaluated to measure how accurately and effectively it recommends relevant books to users.

Model Used

- **Hybrid Recommendation System**

  Combines Collaborative Filtering and Content-Based Filtering to generate accurate recommendations.

Statistical Tools Used

- **RMSE (Root Mean Square Error)**

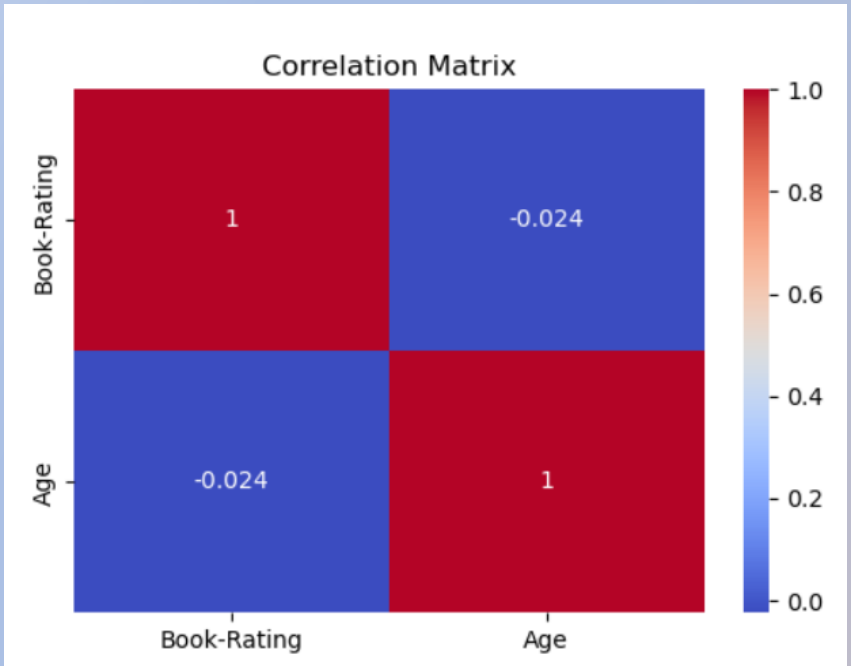
  Measures prediction error between actual and predicted ratings.
- **MAE (Mean Absolute Error)**

  Measures the average prediction error.
- **Precision**

  Measures how many recommended books are relevant.
- **Recall**

  Measures how many relevant books are successfully recommended.

Output :




```
========== CORRELATION ANALYSIS ==========

Correlation between Age and Book Rating:
-0.024190945598175015
```

Explanation

- The **Correlation Matrix** is used to check whether **Age** and **Book Rating** are related.
- Correlation values range from **-1 to +1**:
  - **+1** → Strong positive relationship
  - **-1** → Strong negative relationship
  - **0** → No relationship

Observation

- The correlation value is -0.024, which is very close to 0.
- This means there is almost no relationship between a user's age and the ratings they give to books.

Result

- The weak correlation (-0.024) indicates that age has little impact on book ratings. Hence, recommendations are generated mainly using user preferences and rating patterns instead of age.

Output Explanation

- The system generated 229,956 predicted ratings for user-book combinations.
- Recommendation quality was evaluated using RMSE (3.5040), MAE (2.8168), Precision (0.048), and Recall (0.0253).

Observation

- The model successfully generated a large number of predictions.
- Precision and Recall values indicate that the system can identify relevant books, while some relevant books are still missed.

Result

- The recommendation system was successfully evaluated for accuracy and recommendation quality.
- The results show that the system can generate meaningful book recommendations, with scope for further improvement through model tuning and additional data.

```
Predictions Found Successfully!
Total Predictions: 229956

====================================
PRECISION & RECALL EVALUATION
====================================
Average Precision: 0.048
Average Recall: 0.0253

====================================
INTERPRETATION
====================================
Precision measures how many
recommended books were actually relevant.

Recall measures how many
relevant books were successfully recommended.
```

```
==============================================
MODEL EVALUATION
==============================================
RMSE Score : 3.5040
MAE Score  : 2.8168
```

Conclusion:

- The Hybrid Book Recommendation System successfully combines Collaborative Filtering and Content-Based Filtering to improve recommendation performance.
- The system provides personalized book recommendations based on user preferences and book characteristics.
- Dataset optimization techniques improved the efficiency and scalability of the recommendation process.
- Evaluation metrics (RMSE, MAE, Precision, and Recall) indicate that the system is effective in recommending relevant books to users.

THANKYOU